



I SEMESTER M.C.A

END SEMESTER EXAMINATIONS, DECEMBER 2023

SUBJECT: OBJECT ORIENTED PROGRAMMING [MCA 4121]

REVISED CREDIT SYSTEM

(02/12/2023)

Time: 3 Hours

MAX. MARKS: 50

Instructions to Candidates:

- ❖ Answer **ALL** the questions.
- ❖ Missing data may be suitably assumed.

1A. Given a two-dimensional array of double type containing sales particulars of a shop as follows: 5 CO1 L3

Day	Product1	Product2	Product3	Product4
MON	1750.75	233.90	540.00	567.75
TUE	1675.00	135.75	533.00	134.50
WED	233.90	1675.75	233.90	233.90
THU	1750.75	233.90	540.00	567.75
FRI	1675.75	233.90	233.90	1675.00
SAT	135.75	533.00	134.50	233.90

Write a program to perform the following operations:

- i) Find the total daily sales on each day.
- ii) Find the weekly sale of each product
- iii) Find the average weekly sales of shop.

1B. Consider a class named **Book_Info** that stores information about a book, including the book ID (integer), book title (character array), number of copies (integer), and price (float). Define *default*, *parameterized* and *copy constructors* for initializing objects of this class. 3 CO2 L3

1C. Write C++ statements to dynamically allocate memory for a 2-dimensional array of type double with 3 rows and 4 columns. 2 CO2 L3

2A. What are the steps to implement *pure virtual function* and achieve runtime polymorphism? Provide a real-world example to illustrate these concepts. 5 CO3 L2

2B. Write a C++ program that includes a class called **My_Time** with data members for hours, minutes, and seconds. Define a friend function to overload the unary increment operator 'T++' for an object 'T' of the My_Time class, which increments the time by 1 hour. 3 CO2 L3

2C. Mention any two uses of '*this*' pointer. 2 CO1 L2

3A. Generally, the size of a file may be measured in either bytes or kilobytes. Design a data conversion program to convert input file size from kilobytes to bytes and vice versa. In this program, kilobytes 5 CO2 L3

	should be considered as the user-defined data type, and bytes should be represented by the basic type 'long'.			
3B.	Discuss the steps involved in handling <i>user-defined exception</i> .	3	CO4	L2
3C.	Differentiate between <i>structure</i> and <i>union</i> .	2	CO1	L2
4A.	Consider a class called Patient which stores the information about each patient such as: patient id (integer, starting from 1), patient name (string), age (integer), gender(character). Create the above class with necessary input & output functions and store the objects of this class in a file called 'Patient_info.dat'. You need to display the details of a particular patient based on the patient id entered by the user using the concept of <i>random-access file</i> . Write a complete program to implement the solution to the above problem.	5	CO4	L3
4B.	Write a <i>template function</i> that displays the total number of positive and negative numbers in each column of a two-dimensional array. The function should take the array name and its dimensions (of type integer) as arguments. In the main () function, demonstrate the usage of the template function with arrays of type int, float, and double.	3	CO4	L3
4C.	Given two objects of <i>ofstream</i> and <i>ifstream</i> classes, namely, op_file and ip_file respectively. Write statements to perform the following operations: i) To move the get pointer by 7 bytes in the backward direction from the current position. ii) To move the put pointer to the end of the file.	2	CO4	L3
5A.	How do you display the contents of a vector in both forward and backward directions using <i>iterators</i> and <i>reverse iterators</i> ? Explain with the help of a suitable example.	5	CO5	L2
5B.	Create a function <i>check_email(char email[])</i> to check whether the email address ends with manipal.edu or not. In the main() function, declare a function pointer and invoke the <i>check_email</i> function using this function pointer to determine if the given email address meets the specified criteria.	3	CO3	L3
5C.	Briefly explain (with syntax and example) any two member functions of <i>string</i> class.	2	CO1	L2
